

	GLOBAL ECONOMICS GRADE: 11TH TERM 3 – HOMEWORK HYPOTHESIS TEST DESIGN AND DECISION (C1–C7) TEACHER’S NAME: Nicolás López Cuéllar	THIRD TERM 2025–2026
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Learning objective: Design and apply hypothesis tests from real contexts by identifying the parameter, writing hypotheses, selecting the test direction, applying a proportion test and a mean test, interpreting statistical errors, and determining the minimum sample size needed to reject the null hypothesis.	Evaluated criteria: C1: Describes the concept of a statistical hypothesis. C2: Identifies the null and alternative hypotheses in a statistical test. C3: Determines whether a test should be one-tailed or two-tailed. C4: Applies a hypothesis test in a population proportion setting. C5: Applies a hypothesis test in a population mean counting setting. C6: Identifies Type I or Type II error and its consequences. C7: Finds the minimum sample size needed to reject H_0 .
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Submission instructions. This homework must be submitted as a PDF file by email through Outlook. The maximum turn-in date is Tuesday, June 9, 2026.

Criteria assessment

Each criterion is evaluated through the completion of the corresponding task. A criterion is considered passed when the task is completed correctly. Some minor notation mistakes are allowed, as long as they do not change the procedure, the statistical information, the hypotheses, the graph, the calculations, the decision, or the interpretation. For this homework to count as valid and complete, it must include:

- One consistent real-world counting scenario comparing exactly two groups.
- The same groups, counted variable, and binary threshold throughout C1–C7.
- Evidence of the data-obtention process, such as tables, survey forms, observation records, screenshots, database references, or clearly organized notes.
- The required statistical information for both groups: sample size, mean, standard deviation, binary count, and sample proportion.
- All four setup cases for C1–C3: mean-count with Group A as benchmark, mean-count with Group B as benchmark, proportion with Group A as benchmark, and proportion with Group B as benchmark.
- Only one selected benchmark direction for C4–C7.

Critical values used

$$z_{0.95} = 1.645 \quad z_{0.975} = 1.960 \quad z_{0.99} = 2.326$$

For right-tailed tests, reject when $z > z_{\text{critical}}$. For left-tailed tests, reject when $z < -z_{\text{critical}}$. For two-tailed tests, reject when $|z| > z_{\text{critical}}$.

Formula tools

Mean-count test

$$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

Proportion test

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Minimum sample-size tools:

$$n = \left(\frac{z\sigma}{\bar{x} - \mu_0} \right)^2 \quad n = \frac{z^2 p_0(1-p_0)}{(\hat{p} - p_0)^2}$$

Student problem

Create one consistent real-world counting scenario that compares two groups. Possible comparisons include Prom 2025 vs Prom 2026, Group A vs Group B, before vs after, region 1 vs region 2, two companies, two cohorts, or two markets. The counted variable must be numerical. Examples include late arrivals, completed assignments, absences, deliveries, delays, sales, expenses, minutes, errors, visits, or messages.

The same scenario, groups, counted variable, and binary threshold must be used throughout C1–C7.

Required data table format. Show how the data table is organized so that the counting data and the binary transformation can be understood at the same time. Only a few observations are

needed to show the structure of the table. After that, the remaining observations can be summarized with the required statistical information.

#	Group A count	Group A binary	Group B count	Group B binary
1	count _{A,1}	0 or 1	count _{B,1}	0 or 1
2	count _{A,2}	0 or 1	count _{B,2}	0 or 1
3	count _{A,3}	0 or 1	count _{B,3}	0 or 1
⋮	⋮	⋮	⋮	⋮

Required tasks

C1

Organize the statistical information for both groups and both settings.

- Describe the real-world context, the two groups, and the counted variable.
- Define the binary threshold used to transform each count into 0 or 1.
- Show a few sample rows using the required data-table format.
- Summarize the required statistical information for both groups:

$$n_A, \quad \bar{x}_A, \quad \sigma_A, \quad x_A, \quad \hat{p}_A$$

$$n_B, \quad \bar{x}_B, \quad \sigma_B, \quad x_B, \quad \hat{p}_B$$

- Define the population parameters for the mean-count setting and the proportion setting.

C2

Write the null and alternative hypotheses for all setup cases.

- Mean-count setting using Group A as the benchmark and Group B as the test group.
- Proportion setting using Group A as the benchmark and Group B as the test group.
- Mean-count setting using Group B as the benchmark and Group A as the test group.
- Proportion setting using Group B as the benchmark and Group A as the test group.

Use μ for the mean-count setting and p for the proportion setting.

C3

Classify each setup case as right-tailed, left-tailed, or two-tailed.

- For each setup case from C2, state whether the test is right-tailed, left-tailed, or two-tailed.
- Justify the classification using the sign in the alternative hypothesis.
- Draw an approximate rejection-region graph for each setup case. Each graph must match the selected significance level and the correct tail direction.

Benchmark direction for C4–C7

After completing C1–C3, choose only one benchmark direction.

Selected benchmark direction: _____

All work from C4 to C7 must use only this selected direction. Do not complete C4–C7 for the reverse direction.

C4

Apply the hypothesis test in the proportion setting for your selected benchmark direction.

- State the benchmark proportion p_0 , the sample proportion $\hat{p}$, and the sample size n .
- State the significance level and the correct critical value.
- Compute the test statistic:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

- Compare the test statistic with the appropriate critical value and state the decision about H_0 .
- Interpret the decision in the context given.

C5

Apply the hypothesis test in the mean-count setting for your selected benchmark direction.

- a) State the benchmark mean μ_0 , the sample mean $\bar{x}$, the standard deviation σ , and the sample size n .
- b) State the significance level and the correct critical value.
- c) Compute the test statistic:

$$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

- d) Compare the test statistic with the appropriate critical value and state the decision about H_0 .
- e) Interpret the decision in the context given.

C7

Determine the minimum sample size needed to reject H_0 for the selected benchmark direction.

- a) Find the minimum sample size for the proportion setting:

$$n = \frac{z^2 p_0 (1 - p_0)}{(\hat{p} - p_0)^2}$$

- b) Find the minimum sample size for the mean-count setting:

$$n = \left(\frac{z\sigma}{\bar{x} - \mu_0} \right)^2$$

- c) Round each result up to the next whole number.
- d) Verify that each minimum sample size satisfies the rejection condition.

C6

Revisit the initial-sample-size decision and the minimum-sample-size decision for both settings.

- a) For the proportion setting, compare the initial decision and the decision using the minimum sample size found in C7.
- b) For the mean-count setting, compare the initial decision and the decision using the minimum sample size found in C7.
- c) For each case, identify the possible error type: Type I error or Type II error.
- d) Explain the practical consequence of each possible error in the context given.